Student Name: Pham Xuan Hoang
Student ID: 2640009

USTH
VIETNAM FRANCE UNIVERSITY

Information Systems
Labwork 2

Consider this schema:

- `employees(emp_no, birth_date, first_name, last_name, gender)`
- `departments(dept_no, dept_name)`
- `dept_emp(emp_no, dept_no, from_date, to_date)`
- `dept_manager(dept_no, emp_no, from_date, to_date)`
- `titles(emp_no, title, from_date, to_date)`
- `salaries(emp_no, salary, from_date, to_date)`

Write relational algebra expressions for the following queries (use a pen and paper...)

- All info of all employees: $employees$
- All info of all departments: $departments$
- Full name of all employees: $\pi_{first_name,last_name}(employees)$
- Full name of all departments: $\pi_{dept_name}(departments)$
- Full names of employees working in 'ICT' department:

$$\pi_{first_name,last_name}(\sigma_{dept_name=\texttt{'ICT'}}(employees \bowtie dept_emp \bowtie departments))$$

- Full names of male employees working in 'BIO' department:

$$\pi_{first_name,last_name}(\sigma_{dept_name=\texttt{'BIO'} \wedge gender=\texttt{'M'}}(employees \bowtie dept_emp \bowtie departments))$$

- Salaries of female employees working in 'WEO' department

$$\pi_{salary}(\sigma_{dept_name=\texttt{'WEO'} \wedge gender=\texttt{'F'}}(employees \bowtie dept_emp \bowtie departments \bowtie \pi_{emp_no,salary}(salaries)))$$

- Full names of employees who have the same last name as their manager

$$\pi_{first_name,last_name}\Big((employees \bowtie dept_emp) \\ \bowtie_{dept_no=m_dept \wedge last_name=ln_name \wedge emp_no \neq m_no} \\ \rho_{m_no/emp_no,m_dept/dept_no,ln_name/last_name}(\\ \pi_{emp_no,dept_no,last_name}(employees \bowtie dept_manager))\Big)$$

- Full names of employees who was paid more than $100000

$$\pi_{first_name,last_name}\Big(\sigma_{salary>100000}(employees \bowtie \pi_{emp_no,salary}(salaries))\Big)$$

- Names of all departments that have employees paid more than $100000

$$\pi_{dept_name}\Big(\sigma_{salary>100000}(dept_emp \bowtie \pi_{emp_no,salary}(salaries) \bowtie departments)\Big)$$